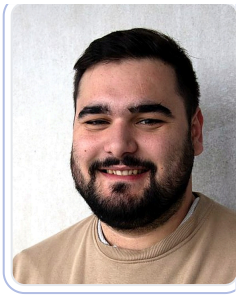
Gustavo Pinho Maia

PhD Researcher in Chemistry · Astrobiology

Email gustavopinho.maia@mnhn.fr ORCID 0000-0001-5314-8816

Website gustavopmaia98.github.io/GustavoPMaia LinkedIn linkedin.com/in/gustavopinhomaia

CiênciaVitae 8212-CEE3-E928 Google Scholar TTVIFykAAAAJ



Credit: IMPMC · Cécile Duflot

RESEARCH INTERESTS

- **Asteroid Gardening and its Effect on Organic Matter.** The goal is to apply continuous mechanical input to molecules already proven to form in ice chemistry and other environments, and see what happens. Do they degrade, assemble into new molecules, or get driven to interact further with mineral structures?
- **Extraterrestrial Ribonucleosides: Possibility or Myth?** Could ribonucleosides really have been made, or, under mechanochemistry, were they broken back down into their components, nucleobases and ribose? They are an essential piece of the RNA World hypothesis.

EDUCATION

2023 – 2027

PhD in Chemistry (Astrobiology)

Ongoing · Universidade de Lisboa · Instituto Superior Técnico

Mechanochemical energy and prebiotic synthesis

My PhD research focuses on understanding how mechanochemical energy (the energy generated from mechanical forces such as grinding, milling, and impact) can drive the synthesis of prebiotic organic molecules. This work explores alternative pathways for the origin of life that don't rely solely on aqueous chemistry.

2020 – 2022

MSc in Chemistry

Concluded · Instituto Superior Técnico

Ribonucleoside stability and prebiotic synthesis

Exogenous delivery of organic molecules could have played a significant role in prebiotic synthesis, based on the detection of RNA nucleobases and ribose in meteorites. However, ribonucleosides were never detected. Meteoroid atmospheric entry conditions are severe and could lead to chemical modifications via mechanochemical reactions. Moreover, oxidation reactions are also plausible, producing reactive species such as carbonate (CO_3^{2-}).

2017 – 2020

BSc in Industrial Chemistry

Universidade da Beira Interior

Optimization of quinoline N-oxide synthesis

Heterocyclic compounds have been studied since the last century. Due to the presence of heteroatoms, these compounds exhibit unique chemical and physical properties, which allow them to interact efficiently with other molecules or even with biological systems.

2013 – 2016

Specialization Level IV as a Mechatronics Technician

Concluded · Escola Profissional de Espinho

Fully automated robot for Rescue Line Competition (RoboCupJunior)

This program focused on advanced mechatronics and robotics skills. The main project was the design and construction of a fully autonomous robot capable of following a line and performing tasks in the Rescue Line competition at RoboCupJunior.

RESEARCH & PROFESSIONAL EXPERIENCE

Jan 2025 – Present

Visiting Scientist
MNHN · IMPMC, France

Collaborative research at MNHN focusing on mechanochemical synthesis, sample analysis, and astrobiology methods.

Nov 2024 – Dec 2024

Visiting Scientist
NASA Goddard Space Flight Center, USA

This study investigated the influence of mechanochemical processes on the extraction and recovery of organic compounds from extraterrestrial materials. Controlled mechanical activation techniques were applied to evaluate their effects on extraction efficiency, molecular integrity, and degradation pathways.

Sep 2024 – Oct 2024

Invited Assistant (University Teacher)
Instituto Superior Técnico, Portugal

As a Guest Teaching Assistant, I delivered and supervised practical laboratory sessions in General Chemistry for first-year undergraduate students in Aerospace Engineering. The practical lectures were designed to reinforce core chemical concepts through hands-on experimentation, with a strong emphasis on laboratory safety, experimental rigour, and data interpretation.

Jun 2021 – May 2022

President of the Fiscal Council
ANEQ, Portugal

Oversaw fiscal compliance, audited financial statements, and coordinated council meetings.

Jul 2020 – Sep 2020

Research Intern
Universidade da Beira Interior

Conducted laboratory research on chemical synthesis under supervision of faculty researchers.

Jun 2020 – Jun 2021

Vice President
ANEQ, Portugal

Assisted the president and managed organizational activities for student chemistry association.

Oct 2019 – Sep 2020

Delegate of Industrial Chemistry Students
Universidade da Beira Interior

Represented the student body, conveyed concerns, and organized class activities and discussions.

Sep 2019 – Dec 2020

Vice-President, UBIQUÍMICA
University of Beira Interior

Managed student projects, coordinated events, and facilitated communication between members.

Aug 2019 – Sep 2019

Factory Worker
Nestlé Portugal

Aug 2018 – Aug 2019

Scientific Committee Member, UBIQUÍMICA
University of Beira Interior

Jun 2016 – Jul 2016

Participant, RoboCup 2016 World Championship (Junior Rescue League)
World robotics championship · Leipzig, Germany

May 2016

Participant, XVI RoboCup Portugal Open
National robotics competition · Bragança, Portugal

2015

Participant, XV RoboCup Portugal Open
National robotics competition · Vila Real, Portugal

Jun 2014 – Jul 2014

Professional Internship
Ramada Aços, S.A.

May 2014

Participant, XIV RoboCup Portugal Open
National robotics competition · Espinho, Portugal

SUPERVISORS

- **José Armando L. da Silva** · Doctoral and Master's advisor · Centro de Química Estrutural · IST
- **Laurent Remusat** · Research supervisor · IMPMC · Muséum national d'Histoire naturelle
- **Jason P. Dworkin** · Research supervisor · NASA · NASA Goddard Space Flight Center
- **Paulo J. S. Almeida** · Bachelor's advisor · Universidade da Beira Interior
- **João Serrano** · Bachelor's co-advisor (then a Master's student) · Universidade da Beira Interior
- **Eng. Carlos Amorim** · Internship coordinator · Professional course · Escola Profissional de Espinho

PEER-REVIEWED PUBLICATIONS

1. **Gustavo P. Maia** et al. (2026). **Effect of asteroid gardening on the exogenous delivery of extraterrestrial organic matter**. *Earth and Planetary Science Letters* 692, 120237. https://doi.org/10.1016/j.epsl.2026.120237
   Keywords: hexamethylenetetramine, montmorillonite, carbonaceous chondrites, interplanetary dust particles, extraterrestrial organic matter, micrometeorite impacts, prebiotic compounds

2. **Gustavo P. Maia**, Catarina Gonçalves, Ana J. Carvalho, Vânia André, Adelino Galvão, Ana P. C. Ribeiro, Pedro F. Pinheiro and José Armando Luísa da Silva (2025). **Mechanochemical Reactivity of Ribonucleosides Mediated by Inorganic Species: Implications for Extraterrestrial Organic Matter Interpretation**. *Applied Sciences* 15(3), 1363. https://doi.org/10.3390/app15031363
   Keywords: nickel, ribonucleosides, aluminum, meteoroid atmospheric entry, extraterrestrial organic matter, mechanochemical degradation, carbonates, planetesimal accretion
   Cited by: 1

3. **Gustavo P. Maia**, José Armando Luísa da Silva, Vânia André and Adelino M. Galvão (2023). **Shock-Induced Degradation of Guanosine and Uridine Promoted by Nickel and Carbonate: Potential Applications**. *Molecules* 28(24), 8006. https://doi.org/10.3390/molecules28248006
   Keywords: uridine, uracil, mechanochemistry, nickel(II) catalysis, guanine, guanosine, extraterrestrial samples, astrobiology Cited by: 2

4. Ana Franco, **Gustavo P. Maia** and José Armando Luísa da Silva (2023). **Da Química Bioinorgânica para a Química Prebiótica: Levar Mais Longe o Legado do Prof. Fraústo da Silva**. *Boletim da Sociedade Portuguesa de Química*. https://doi.org/10.52590/M3.P708.A30002745

CONFERENCE PRESENTATIONS

Nov 2026

Effect of asteroid gardening on extraterrestrial organic matter: implications for the RNA World Hypothesis ORAL
EJC 2026 · CNES, Paris
Oral talk · Maia, Dworkin, McLain, da Silva & Remusat

Aug 2026

Effect of Mechanochemistry on Organic Matter: Implications for Impact Gardening on Asteroid Surfaces ORAL
MetSoc 2026 · Frankfurt
Oral talk · Maia, Remusat, Amano, Viennet, Dworkin, McLain & da Silva

Jul 2026

Mechanochemical Processing of Organic Matter During Asteroid Gardening ORAL
Origins 2026 (ISSOL) · Paris
Oral talk · Maia, Remusat, Amano, Viennet, Dworkin, McLain & da Silva

Oct 2025

The effect of mechanochemistry on the formation of extraterrestrial organic matter ORAL
AbGradE'25 · Lisbon
Oral talk · Maia, Dworkin, da Silva & Remusat

Jul 2025

Effect of mechanochemical events on extraterrestrial organic matter and its implications for exogenous delivery ORAL

BEACON 2025 · Harpa, Reykjavik

Oral presentation · BEACON 2025

Jun 2025

The effect of shock/impacts on the exogenous delivery of organic matter ORAL

IMPMC PhD Day · Paris

Oral talk · IMPMC, Sorbonne Université

Jan 2025

Mechanochemistry of extraterrestrial bodies: Bridging exogenous delivery with prebiotic chemistry ORAL

MNHN · Paris

Oral talk · MNHN, Paris

Jan 2025

Mechanochemistry and astrobiology: mimicking impact reactions ORAL

MNHN · Paris

Oral talk · MNHN, Paris

2025

Solid-state mechanochemistry of extraterrestrial organic molecules: new insights on the exogenous delivery of organic matter POSTER

EANA 2025 · Lisbon · Poster Award

Award-winning poster · EANA 2025

2025

Mechanochemistry of small bodies and the evolution of extraterrestrial organic matter ORAL

IPGP "Small Bodies Day" 2025 · Paris

Oral talk · IPGP, Paris

2024

Impact events of extraterrestrial bodies: the bridge between exogenous delivery and the origin of life ORAL

NInTec Science Days 2024 · Lisbon

Oral talk · NInTec Science Days 2024

2024

Shock-Impact Events and the Origin of Life: a mechanochemical approach POSTER

IST PhD Open Days 2024 · Lisbon

Poster · PhD Open Days 2024

Aug 2022 – Sep 2022

Ribonucleoside stability and prebiotic synthesis ORAL

8th EuChemS Chemistry Congress · CCL, Lisbon

Oral talk · 8th EuChemS Congress

2022

Ribonucleosides stability and prebiotic synthesis POSTER

CQE Days 2022 · Lisbon · P083

Poster · CQE Days 2022

AWARDS & DISTINCTIONS

2026

Excellent Teachers 2024/2025 · Excellence in Teaching

Pedagogical Council · Instituto Superior Técnico, Universidade de Lisboa

Diploma of teaching excellence for teaching General Chemistry (LEAN; LEAer21) in the 2024/2025 academic year.

2025

EANA 2025 Poster Award

EANA 2025 · Lisbon, Portugal

Poster Award at the European Astrobiology Network Association annual conference, recognising research on mechanochemistry and the prebiotic synthesis of organic molecules.

2017

Diploma of Merit Recognition

Rotary Club de Espinho

Merit recognition diploma awarded for outstanding academic performance in 2015/2016 at the Escola Profissional de Espinho (CEPROF).

2016

Best Presentation · RCJ Rescue Maze

RoboCup 2016 · Leipzig, Germany

Awarded to team ROB07s (Escola Profissional de Espinho) at the RoboCup world championship for the best presentation in the RoboCup Junior Rescue Maze league.

FUNDING

2023

Mechanochemical energy and prebiotic synthesis

Doctoral Grant · Fundação para a Ciência e a Tecnologia (FCT) · 2023.01099.BD

DOI: 10.54499/2023.01099.BD

Jun 2023 – Nov 2023

A Gold Nanosensor for Single Molecule Detection of RNA Biomarkers in a Miniaturized Microscope

Research Fellow · Centro de Química Estrutural, Universidade de Lisboa

2020

Centro de Química Estrutural

Research Unit Funding · Fundação para a Ciência e a Tecnologia (FCT) · UIDB/00100/2020

DOI: 10.54499/UIDB/00100/2020

COURSES & TRAINING

Jun 2026

Building the Hard Rocky Planets — From Mercury to the Asteroid Belt

Summer School · Institut Origines (Aix-Marseille Université) & Institut de Physique du Globe de Paris (IPGP) · Marseille, France

Jun 2026

Astrobiology Introductory Course (Virtual)

Course · Rencontres Exobiologiques pour Doctorants (RED) · Le Teich, France · Online

Mar 2025

Astrobiology Introductory Course

Course · Rencontres Exobiologiques pour Doctorants (RED) · Le Teich, France

Feb 2025

Recommendations for Publishing in International Journals

Formation · Biblioteca do Instituto Superior Técnico (BIST) · Online

TECHNIQUES & SOFTWARE

Analytical techniques	Chromatography & MS, NMR, XRD, FTIR, EA, irMS, SEM · EDX
Instruments & data	TopSpin, MestReNova, OpenLab, MassLynx & TargetLynx, OPUS, Profex, ACD/Labs
Synthesis & modelling	Mechanochemistry, Organic Synthesis, Ab Initio
Chemistry & computation	Python, R, MATLAB, ChemDraw, Chemcraft
Engineering & electronics	Arduino, PICAXE, Multisim, SOLIDWORKS, AutoCAD, SketchUp
Productivity & code	Office 365, VS Code, GitHub

MEDIA & RECOGNITION

2026

Recognised among the “Excellent Teachers 2024/2025” at Instituto Superior Técnico (Universidade de Lisboa), for teaching in the General Chemistry laboratories.

Sep 2025

Featured in Maré Viva (Jornal Regional de Espinho): “O percurso improvável de Gustavo Maia levou-o de Ovar até à NASA”, a profile tracing the path from Ovar and the Escola Profissional de Espinho, through world robotics competitions, to a NASA laboratory.

Dec 2024

Interviewed by Praça Pública: “Gustavo Maia aprofunda investigação no Laboratório Analítico de Astrobiologia da NASA”.

Nov 2024

Interviewed by Diferencial, the IST student journal: “Portanto, a Vida foi criada a partir de surras.”

Generated from the website on 13 Nov 2026.